



Scout Boom

Deployable Camera Mechanism

The Scout Boom is a deployable camera mechanism designed for capturing visual inspection data and "selfies" of the satellite in orbit.

It features an integrated flex PCB running along the boom length, eliminating complex wire harnesses and reducing snag risks. The system uses stored strain energy to deploy a bistable composite boom rolled compactly against the CubeSat.

SPECIFICATIONS

Deployment	Stored Strain Energy
Form Factor	1U+ Compatible
Length	Up to 1.5 m
System Mass	< 150g (w/ Cam)
Tip Mass	40g (at 70cm)
Camera	5 MP (Typical)
Interface	MIPI CSI-2 / USB
Power	3.3V / 150mW

KEY FEATURES

Integration	Embedded Flex PCB
Actuation	Burn Wire (HDRM)
Feedback	Limit Switch
TRL	4 (Prototype)